



**Politecnico
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Nuclear engineering lab and advanced heat transfer problems

Sustainable Nuclear Energy

TPMS heat exchanger

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Nomenclature

Latin Symbols

$\dot{m}$	Mass Flow Rate [kg/s]
$\dot{Q}$	Heat Transfer Rate/Thermal Power [W]
A	Heat Transfer Surface Area [m ²]
A_{in}	Fluid Inlet Cross-Sectional Area [m ²]
A_{wet}	Wetted Surface Area [m ²]
D_h	Hydraulic Diameter [m]
e_a	Relative Error [-]
F_s	Safety factor used in the GCI evaluation [-]
f_{ext}	Continuum-Extrapolated Value
h	Convective Heat Transfer Coefficient [W/(m ² K)]
L_{tot}	Total Lenght [m]
T	Temperature [K]
u_D	Experimental Measurement Uncertainty [-]
u_{input}	Input Uncertainty [-]
u_{num}	Numerical Discretization Uncertainty [-]
u_{val}	Validation Uncertainty [-]
V_{void}	Void Volume [m ³]
V_{sup}	Superficial Velocity [m/s]
f	Friction Factor [-]
G	Goodness Factor [-]
j	Colburn Factor [-]
Nu	Nusselt Number [-]
p	Order of Convergence [-]
Pr	Prandlt Number [-]
r	Grid Refinement Ratio [-]
Re	Reynolds Number [-]
St	Stanton Number [-]

Greek Symbols

ΔP	Pressure Drop [Pa]
ΔT_{lm}	Logarithmic Mean Temperature Difference [K]
Δ	Difference/Variation Operator [-]
δ_{model}	Modeling Error [-]
μ	Dynamic Viscosity [Pa · s]
ρ	Density [kg/m ³]
ε	Medium Porosity [-]



Acronyms

AM	Additive Manufacturing
CHT	Conjugate Heat Transfer
DAQ	Data Acquisition
DLP	Digital Light Processing
FKS	Fischer-Koch S
FOM	Figure of Merit
FVM	Finite Volume Method
GCI	Grid Convergence Index
HX	Heat Exchanger
L-PBF	Laser Powder Bed Fusion
PCM	Phase Change Material
RANS	Reynolds-Averaged Navier-Stokes
SLA	Stereolithography
SLM	Selective Laser Melting
TPMS	Triply Periodic Minimal Surface
UV	Ultraviolet



1 Introduction

In recent years, the power generation sector has faced an increasing demand for enhanced heat extraction capabilities. For instance, in nuclear fusion applications such as the ITER, plasma-facing components must be capable of withstanding and efficiently extracting heat fluxes on the order of 10 MW/m^2 .

The fundamental mechanism of convective heat transfer is governed by Newton's law of cooling:

$$\dot{Q} = h \cdot A \cdot \Delta T \quad (1.1)$$

where $\dot{Q}$ represents the heat transfer rate, h express how much the fluid is efficient to exchange heat with the solid surface, A is the heat transfer surface area, and ΔT is the temperature difference between the solid surface and the bulk fluid.

The fundamental objective of thermal management is to maximize the heat transfer rate, $\dot{Q}$. To enhance the overall thermal performance of a device, engineers can theoretically intervene on the variables of heat transfer equation (1.1). However, each approach presents specific engineering challenges and operational limitations. Increasing ΔT is often practically unfeasible, as it is strictly constrained by the thermodynamic cycle requirements and the maximum allowable operating temperatures of the structural materials. Enhancing the convective heat transfer coefficient typically requires disrupting the thermal boundary layer—which acts as an insulator—by inducing turbulence, flow separation, or secondary flows through active or passive enhancement methods. While thermally effective, this approach inevitably incurs a severe penalty in terms of ΔP and required pumping power, thus reducing the overall thermodynamic efficiency of the system due to viscous entropy generation. Finally, increasing the heat transfer area is conventionally achieved using extended surfaces, such as fins, or stochastic porous media. Nevertheless, this introduces geometric drawbacks, including increased weight, larger volumetric footprints, and higher hydraulic resistance, often accompanied by the formation of stagnant fluid regions (dead zones).

To satisfy increasingly stringent thermal management requirements, engineers have developed highly sophisticated heat exchange architectures. Among these, Triply Periodic Minimal Surfaces have recently emerged as a revolutionary topological solution.

1.1 Triply Periodic Minimal Surfaces

Formulated via 3D trigonometric level-set equations, Triply Periodic Minimal Surfaces are characterized by a zero mean curvature and continuous tortuosity. These lattices inherently provide an exceptionally high surface-area-to-volume ratio, establishing them as an optimal topology for compact heat sinks. Furthermore, their continuous curvature induces persistent flow separation and boundary layer collisions, which substantially enhance the local convective heat transfer coefficient. Unlike conventional porous media, TPMS architectures achieve this superior thermal performance without generating severe stagnant dead zones, thereby maintaining a manageable pressure drop penalty.

The theoretical framework of TPMS relies on two fundamental geometric principles:

- **Minimal Surface:** mathematically, it is defined as a surface that locally minimizes its area. Its defining geometric trait is a mean curvature identically equal to zero at every single point.

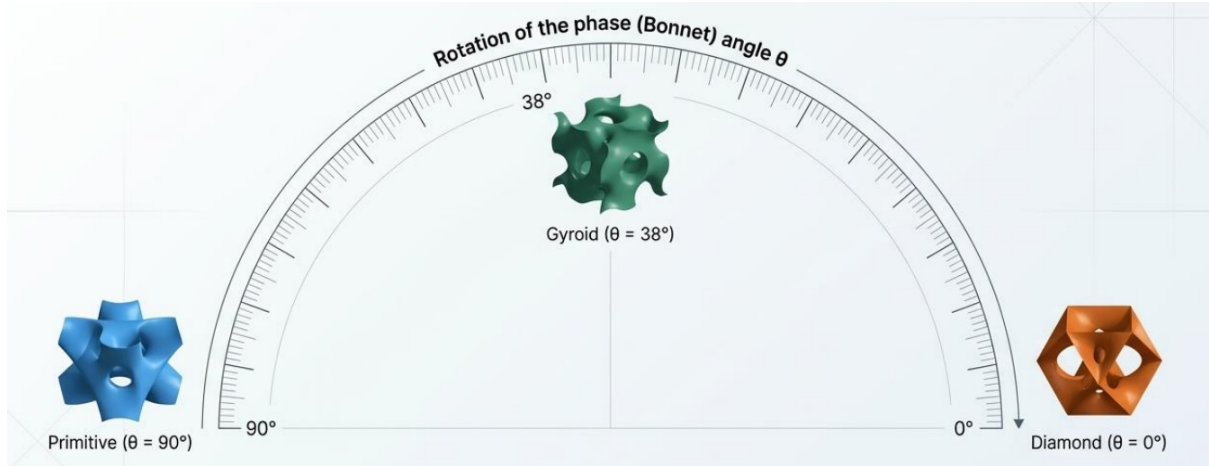


Figure 1.1: Popological transformation dictated by the Weierstrass-Enneper phase rotation

- **Triply Periodic:** this property indicates spatial periodicity along all three Cartesian axes (X, Y, and Z). The topology is generated through level-set equations combining trigonometric functions (sine and cosine), which define a fundamental unit cell that repeats infinitely in three-dimensional space without ever self-intersecting.

The three fundamental Triply Periodic Minimal Surface topologies are categorized by their mathematical formulation:

- **Primitive (P-surface):** As the most basic architecture, it is defined by a single trigonometric function per term.
- **Gyroid (G-surface):** Exhibiting increased complexity, this structure relies on the product of two distinct trigonometric functions within each term.
- **Diamond (D-surface):** Characterized as the most geometrically intricate of the primary topologies, it is generated through the product of three different trigonometric functions.

These specific configurations are unified under the same mathematical "Associate Family". Possessing an identical conformal framework, they can be continuously morphed into one another through a topological transformation dictated by the Weierstrass-Enneper phase rotation. This isometric deformation process is strictly controlled by the phase angle θ , conventionally designated as the Bonnet angle

Beyond the primary topologies, it is possible to generate advanced architectural configurations, such as:

- **I-WP and Fischer-Koch S:** Both structures belong to the Diamond-related family, sharing a cubic primitive lattice. The I-WP topology exhibits lower symmetry, generating two morphologically distinct, non-self-complementary labyrinths. Conversely, the FKS structure is highly intricate, providing exceptionally elevated surface-area-to-volume ratios while maintaining perfect domain symmetry; however, this architecture incurs significant pressure drop penalties owing to its highly constricted channel dimensions.
- **Neovius:** This topology shares the identical Body-Centered Cubic lattice with the Primitive surface, yet it is characterized by fundamentally asymmetric flow channels.



- **SplitP:** Derived via the partial symmetry breaking of the Primitive surface, this architecture is highly non-self-complementary. Consequently, the two resulting labyrinth domains possess drastically different morphological shapes and hydraulic impedances. When designing a solid layout, the engineer must actively determine which specific region to fill with solid material, as one channel exhibits a markedly higher flow resistance than the other. It serves as an extremely effective configuration for heat sink applications.
- **Lidinoid:** Functioning as a non-chiral analogue to the Gyroid, this structure presents a critical design constraint: if the porosity exceeds 50%, the geometry can transition into a tricontinuous topology. This morphological shift leads to the emergence of a third, completely disconnected fluid cavity. Although this third channel is detrimental to continuous flow percolation in standard convective heat exchangers, it can be strategically leveraged by filling it with Phase Change Materials to develop advanced latent thermal energy storage systems.

These architectures can be manufactured in two distinct configurations, each tailored for specific thermal management applications:

- **Solid layout:** One of the two labyrinth regions is entirely filled with solid material, leaving the other empty as a single fluid domain. This topology provides high structural rigidity and is the preferred choice for heat sinks (Fig.1.2).
- **Sheet layout:** This configuration is created by displacing the iso-surface slightly upward and downward, resulting in a thin separating solid wall of constant finite thickness. This leaves both labyrinths empty, yielding two separate fluid domains. Because a heat exchanger inherently requires two different fluid channels, the sheet configuration is mandatory for these devices (Fig. 1.3).

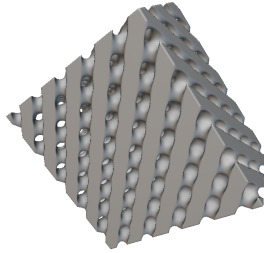


Figure 1.2: TPMS solid layout

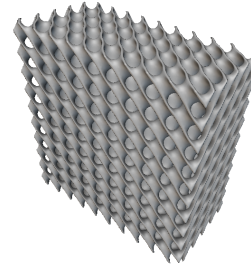


Figure 1.3: TPMS sheet layout

1.2 Additive Manufacturing

The theoretical foundation of TPMS was established in 1865 by the German mathematician Hermann Amandus Schwarz. However, the engineering relevance of these geometries has only recently surged, driven by the widespread diffusion of additive manufacturing technologies. Prior to AM, fabricating these inherently tortuous architectures was practically impossible, as traditional subtractive manufacturing methods (e.g., milling or casting) cannot physically create infinite internal labyrinths characterized by zero mean curvature. Additive manufacturing acts as the technological catalyst that transitions



TPMS from mathematical concepts to physical engineering realities. The primary AM technologies employed for TPMS heat exchangers include:

- **Metal AM:** Selective Laser Melting and Laser Powder Bed Fusion enable the use of materials with excellent thermal conductivity, such as pure Copper or AlSi10Mg. These technologies are the standard for manufacturing operational heat sinks and compact heat exchangers.
- **Resin AM:** Stereolithography provides ultra-high geometric resolution, making it ideal for prototyping and cold fluid dynamics visualization tests.

Despite its potential, the layer-by-layer 3D printing process introduces severe manufacturing constraints and defects that critically impact thermal-hydraulic performance:

- **Overhanging Layers:** Printing complex TPMS topologies requires careful calibration of the printing angle to ensure the structure remains self-supporting without the need for internal support structures.
- **Internal Defects:** Manufacturing flaws, such as lack of fusion or keyholing, inherently lower the effective thermal conductivity of the solid matrix compared to the theoretical properties of the bulk material.
- **Surface Defects and Roughness:** The layer-by-layer deposition leads to staircase effects, and non-fused powder frequently adheres to the lower, unsupported surfaces (downskin). This unplanned surface roughness drastically alters the fluid friction, generating pressure drops that can exceed ideal CAD-based analytical predictions by up to 69%.

Nevertheless, Additive Manufacturing represents a crucial technological enabler, as it provides the most effective means to fabricate highly complex and tortuous architectures.

2 Workflow and Project Aims

Having established the theoretical advantages of TPMS architectures in enhancing convective heat transfer, it is imperative to validate these capabilities within a concrete engineering framework. Consequently, the following sections detail the computational design and evaluation of a specific TPMS-based heat exchanger.

2.1 Project Aims and Constraints

The primary objectives of this study are the design and experimental testing of a TPMS HX, followed by the validation of the computational framework utilized for the preliminary analysis of various lattice topologies and dimensions. Crucially, the overarching goal is to achieve maximum thermo-hydraulic efficiency while strictly minimizing structural material volume to optimize the compactness and cost-effectiveness of the additive manufacturing process.

Prior to the parametric design and thermo-hydraulic optimization, the physical domain is bounded by strict geometric envelope limits and additive manufacturing constraints:

- **Maximum spatial envelope:** $65 \times 55 \times 125$ mm (Fig. 2.1).
- **Minimum structural thickness:** 1 mm (Fig. 2.2).

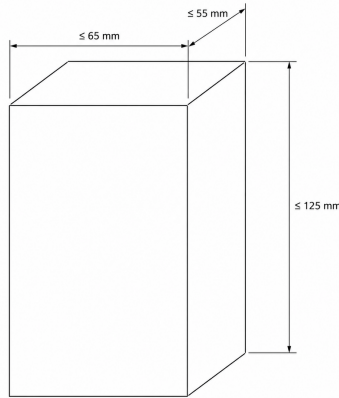


Figure 2.1: Maximum allowable spatial envelope for the heat exchanger

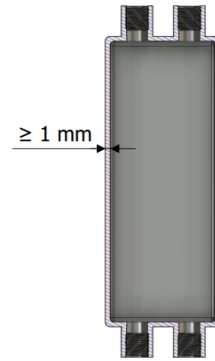


Figure 2.2: Minimum wall thickness dictated by manufacturing limits

- **Inter-manifold spacing:** 20 mm (Fig. 2.3).
- **Manifold port diameter:** 1/2 inch (Fig. 2.4).

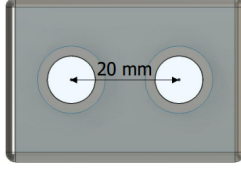


Figure 2.3: Required geometric spacing between adjacent manifolds

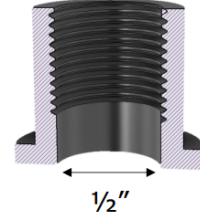


Figure 2.4: Standardized internal diameter of the manifold ports

- **Countercurrent flow:** To enhance convective heat transfer, the streams follow a forced cross-counter-current path, maximizing fluid residence time and core mixing (Fig. 2.5).

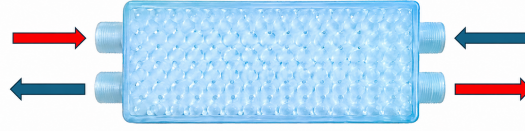


Figure 2.5: Cross-counter-current flow

2.2 Workflow

To achieve the defined thermo-hydraulic design objectives while adhering to the physical, manufacturing, and spatial constraints, a structured multi-stage methodological workflow was established.

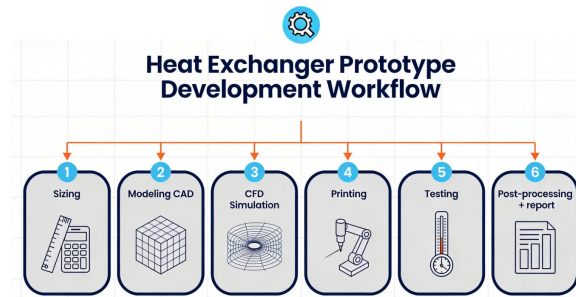


Figure 2.6: Methodological workflow adopted for the design of the TPMS heat exchanger.

As outlined in the workflow sequence (Figure 2.6), the process consists of the following sequential stages:

1. **Parametric Sizing & Sensitivity Analysis:** To maximize computational efficiency while thoroughly evaluating the design space, the numerical investigation was conducted following a systematic two-stage strategy on representative computational sub-domains:
 - **Lattice Topology Benchmark:** Under identical macroscopic boundary conditions and a fixed cubic unit cell volume ($20 \times 20 \times 20$ mm), four distinct



candidate TPMS architectures (Diamond, SplitP, Lidinoid and Gyroid) were systematically simulated and compared.

- **Diamond Cell Sizing & Aspect Ratio Sweep:** Based on prior literature predictions highlighting its superior convective heat transfer performance, the Diamond topology was selected a priori for an extensive geometric sensitivity study. The analysis investigated both isotropic and anisotropic cell configurations, specifically evaluating $10 \times 10 \times 10$ mm, $15 \times 15 \times 15$ mm, and $15 \times 20 \times 20$ mm cell envelopes.
- **Flow Rate and Reynolds Number Sensitivity:** All topological and dimensional test configurations were evaluated across multiple operational mass flow rates. To preserve dynamic similarity and ensure physical consistency across the comparative sweep, the mass flow rate ratio between the hot and cold fluid streams ($\dot{m}_{hot}/\dot{m}_{cold}$) was strictly held constant according to the baseline design constraints.

Since the resin acts as a thermal insulator, the TPMS wall thickness was initially set to the minimum feasible value (1 mm); however, due to additive manufacturing constraints, it was subsequently increased to 1.8 mm to ensure structural integrity.

2. **CAD Modeling:** The geometric generation of the computational domain was partitioned across two distinct software platforms according to functional complexity:
 - **nTopology:** Employed for the implicit mathematical synthesis of the TPMS lattice core, generating the zero-mean-curvature sheet architectures via trigonometric level-set formulations.
 - **SolidWorks:** Utilized for the parametric design of the macroscopic fluid containment, including the outer shell housing and the inlet/outlet manifold.
3. **CFD Simulation and Solution Verification:** Following the generation of the TPMS core geometry in nTop and its subsequent export into STAR-CCM+, steady-state laminar CHT simulations are performed across the coupled solid and fluid domains. The fluid flow and heat transport are governed by the full, incompressible, steady-state Navier-Stokes and energy equations, solved simultaneously using a segregated finite-volume approach without turbulence modeling. Near-wall prism layers cells are implemented to resolve the steep spatial velocity and temperature gradients induced by the no slip condition.

To ensure that spatial discretization errors do not contaminate the physical solution, a rigorous Solution Verification is carried out in compliance with the ASME V&V 20-2009 standard. The GCI methodology, based on Generalized Richardson Extrapolation, is systematically applied across at least three geometrically similar, refined mesh levels ($r > 1.3$). The observed order of convergence (p) and the continuum-extrapolated values (f_{ext}) are extracted to compute the numerical uncertainty on the finest mesh, isolating numerical discretization errors prior to performance evaluation and experimental validation.
4. **Additive Manufacturing of the Prototype:** The CAD model of the HX was exported from nTop in *.obj* format to ensure seamless integration with the 3D printer's slicing software. The prototype was fabricated via SLA. While this vat-photopolymerization technique provides superior dimensional accuracy and high



surface resolution, both critical for resolving intricate TPMS topologies, the resulting photopolymer exhibits low thermal conductivity. Post-fabrication, the green part¹ was thoroughly washed in an organic solvent bath to remove uncured resin residues that could obstruct the narrow TPMS flow channels. Finally, the component underwent UV post-curing to complete the cross-linking of the polymer matrix, thereby establishing its nominal mechanical strength and structural rigidity.

5. **Experimental Testing:** The manufactured prototype was experimentally tested in a laboratory flow loop using air as the working fluid. A Python script was deployed to acquire, log, and export the inlet and outlet temperatures of both channels into a *.csv* data file.
6. **Post-Processing & Model Validation:** The experimental datasets were compared against CHT CFD simulations of the full-scale heat exchanger, executed under matching boundary conditions to assess model validity.

¹In AM, the term *green part* refers to the newly fabricated component immediately after the photopolymerization process, prior to post-curing or secondary thermal treatments required to achieve full polymer cross-linking and nominal mechanical strength.

3 Preliminary Design and Morphological Analysis

This chapter presents the iterative design methodology and morphological steps required to develop the final CAD geometry, which was subsequently fabricated via additive manufacturing and experimentally tested.

3.1 Preliminary Design

Prior to the additive manufacturing of the HX, several key design parameters must be defined: the unit cell size, the number of unit cells, the TPMS topology, and the wall thicknesses (both external structural walls and internal TPMS sheets).

- **Wall thickness:** Because minimizing resin consumption is a primary design objective, the external casing thickness is fixed at the minimum allowable fabrication limit of 1 mm. Similarly, the initial TPMS sheet thickness is set to this minimum threshold to mitigate the poor thermal conductivity of the resin, thereby reducing conductive thermal resistance and enhancing overall heat transfer.
- **TPMS topology and domain dimensions:** Since the convective heat transfer rate (Eq. 1.1) depends directly on the available surface area, the preliminary strategy seeks to maximize this parameter by utilizing the maximum allowable HX envelope shown in Fig. 2.1. Although local heat transfer area varies across topologies, this preliminary stage focuses primarily on the global heat transfer performance. Consequently, all preliminary TPMS models are simulated within a $40 \times 40 \times 100 \text{ mm}^3$ domain, striking a balance between computational cost and geometric consistency across test configurations. Consequently, the unit cell size and the total number of cells are determined based on the optimal trade-off identified from the simulation results.

A parametric study was conducted to evaluate the influence of different TPMS topologies and unit cell sizes. As illustrated in Fig. 3.1, the investigated topologies were initially tested under an identical unit cell size. Furthermore, based on evidence from the scientific literature highlighting its high heat transfer capabilities, the Diamond topology was selected a priori for an extended evaluation across multiple unit cell dimensions.

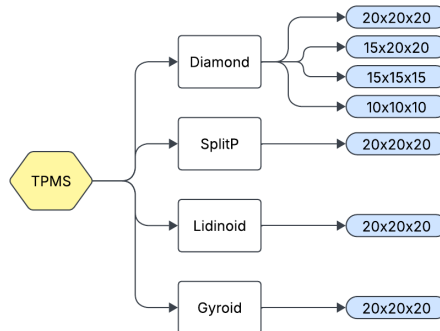


Figure 3.1: Overview of the investigated TPMS topologies and unit cell configurations



While the nominal mass flow rate ratio between the hot and cold streams ($\dot{m}_{\text{hot}}/\dot{m}_{\text{cold}}$) was maintained constant based on the baseline design, the total mass flow rate was scaled by factors of $2\times$, $5\times$, and $20\times^2$. For the initial test matrix, the Reynolds number remained sufficiently low to justify a laminar flow formulation; conversely, for the highest mass flow rate conditions, the Reynolds number increased significantly, necessitating a transition to a turbulence model.

3.2 Performance Evaluation Criteria

To evaluate and rank the thermal-hydraulic performance of the investigated TPMS architectures, two primary FOMs were employed: the Colburn factor eq and the Flow Area Goodness Factor. Because TPMS lattices behave as complex architected porous media, all dimensionless numbers and hydraulic parameters were formulated taking into account the medium porosity and the superficial flow velocity.

The Colburn factor provides a dimensionless measure of convective heat transfer effectiveness across varying flow regimes, independent of fluid thermophysical properties:

$$j = \text{St} \cdot \text{Pr}^{2/3} = \frac{\text{Nu}}{\text{Re} \cdot \text{Pr}^{1/3}} \quad (3.1)$$

Under the porous medium formulation, the pore-level Reynolds number is evaluated incorporating the medium porosity:

$$\text{Re} = \frac{\rho u D_h}{\mu \varepsilon} \quad (3.2)$$

The characteristic hydraulic diameter of the lattice is defined as

$$D_h = 4V_{\text{void}}/A_{\text{wet}} \quad (3.3)$$

To quantify the hydraulic penalty and flow resistance across the porous matrix, the friction factor is defined using the superficial velocity that take in to account the porosity:

$$f = \frac{\Delta P \cdot D_h}{2\rho L_{\text{tot}} V_{\text{sup}}^2} \quad (3.4)$$

Finally, the comprehensive trade-off between heat transfer enhancement and required pumping power is evaluated through the Flow Area Goodness Factor:

$$G = \frac{j}{f} \quad (3.5)$$

Derived from the Reynolds analogy, this figure of merit assesses the thermodynamic efficiency of the lattice, where higher j/f values reflect superior thermal augmentation for a given pumping power penalty.

²The nominal design flow rate was intentionally superseded in this preliminary phase to induce transitional and turbulent flow regimes, thereby amplifying performance discrepancies among the investigated TPMS topologies, as preliminary runs exhibited negligible differentiation under low-Reynolds-number conditions.



3.3 Results and Discussion

As outlined in Section 3.1, a parametric numerical campaign was conducted across multiple TPMS topologies and unit cell sizes. A reduced computational domain was employed to optimize computational costs without compromising accuracy.

Multiplier	Geometry	Cell size	j_{min}	$j_{min} (por)$	G_{min}	$G_{min} (por)$
2	Diamond	20x20x20	3.92E-02	1.65E-02	3.29E-03	1.86E-02
2	Diamond	15x20x20	3.53E-02	1.38E-02	1.79E-03	1.18E-02
2	Diamond	15x15x15	2.64E-02	9.08E-03	1.81E-03	1.52E-02
2	Diamond	10x10x10	1.67E-02	5.69E-03	9.52E-04	8.19E-03
2	SplitP	20x20x20	2.88E-02	6.77E-03	1.46E-03	1.29E-02
2	Lidinoid	20x20x20	1.97E-02	8.04E-03	2.12E-04	4.22E-03
2	Gyroid	20x20x20	4.93E-02	1.98E-02	3.07E-03	1.90E-02
5	Diamond	20x20x20	3.95E-02	1.66E-02	5.58E-03	3.15E-02
5	Diamond	15x20x20	3.57E-02	1.39E-02	2.85E-03	1.87E-02
5	Diamond	15x15x15	5.74E-03	1.98E-03	5.17E-04	4.36E-03
5	Diamond	10x10x10	2.07E-02	7.06E-03	1.35E-03	1.16E-02
5	SplitP	20x20x20	2.87E-02	6.55E-03	2.23E-03	1.67E-02
5	Lidinoid	20x20x20	2.23E-02	6.80E-03	2.72E-04	5.41E-03
5	Gyroid	20x20x20	1.94E-01	7.81E-02	1.79E-03	1.11E-02
20	Diamond	20x20x20	4.01E-02	1.69E-02	9.68E-03	5.46E-02
20	Diamond	15x20x20	3.57E-02	1.40E-02	4.61E-03	3.03E-02
20	Diamond	15x15x15	2.95E-02	1.02E-02	4.39E-03	3.70E-02
20	Diamond	10x10x10	2.10E-02	7.15E-03	2.58E-03	2.22E-02
20	SplitP	20x20x20	2.88E-02	6.47E-03	3.28E-03	1.99E-02
20	Lidinoid	20x20x20	2.21E-02	6.76E-03	3.87E-04	7.70E-03
20	Gyroid	20x20x20	4.86E-02	1.95E-02	7.17E-03	4.44E-02

Table 3.1: Summary of preliminary CFD simulation results across the investigated TPMS topologies and flow rates.

The preliminary thermo-hydraulic results, summarized in Table 3.1, highlight distinct performance behaviors depending on the evaluation metric employed:

- Across all investigated mass flow rates, the Gyroid topology consistently exhibits the highest Colburn factor, demonstrating superior convective heat transfer performance per unit of surface area.
- Conversely, when hydraulic penalties and pressure losses are accounted for through the Flow Area Goodness Factor, the Diamond configuration clearly outperforms all competing topologies, achieving the most favorable balance between heat transfer enhancement and pumping power dissipation.

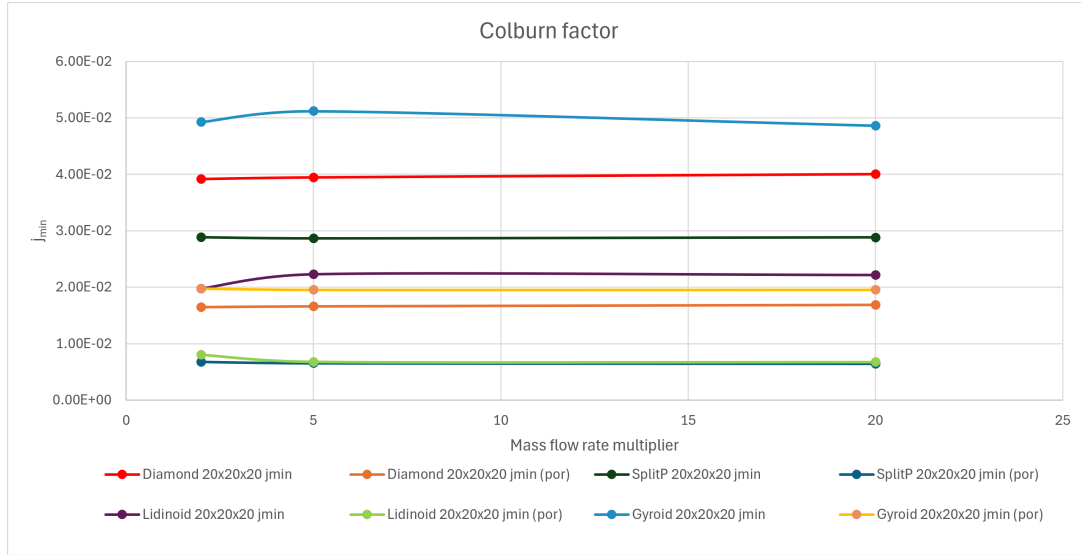


Figure 3.2: Comparison of the Colburn factor for different TPMS topologies, evaluated with and without porosity correction.

It is worth emphasizing that all performance metrics were evaluated using two distinct mathematical formulations: standard channel flow definitions and porous medium formulations explicitly incorporating the medium's porosity. Evaluating both sets of equations enabled a direct comparative analysis, assessing the impact of pore-level velocity scaling and void fraction on the resulting dimensionless numbers.

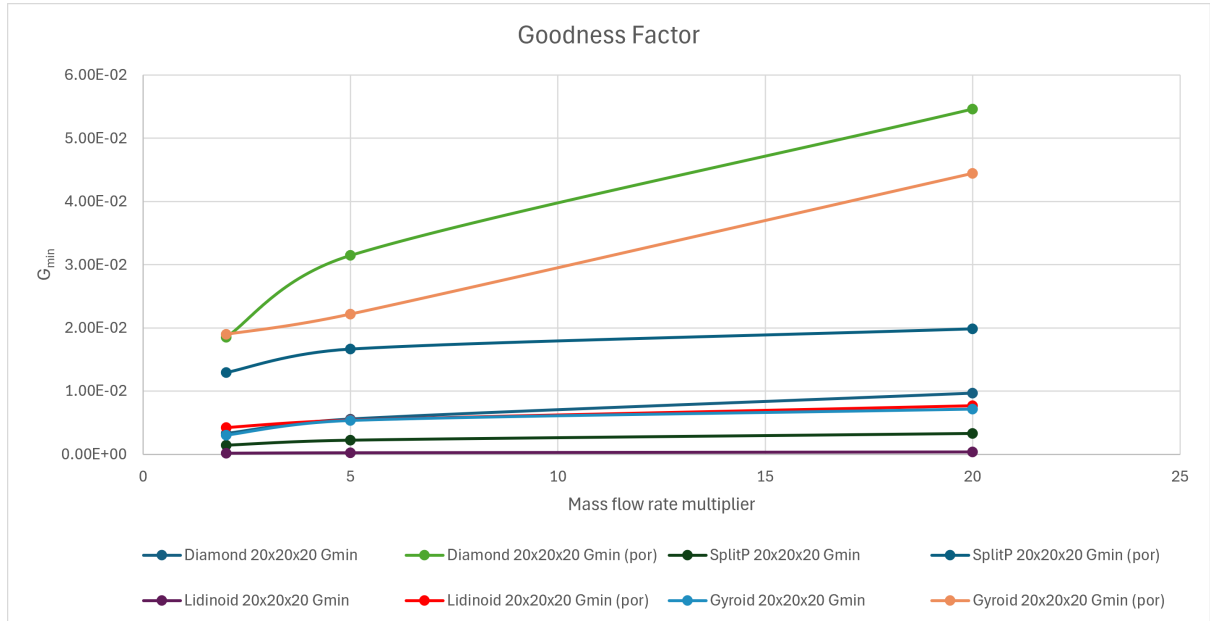


Figure 3.3: Goodness factor comparison across different TPMS topologies as a function of the scale multiplier.

While the difference in the Colburn factor across the analyzed topologies remains nearly constant (Figure 3.2), the Flow Area Goodness Factor (Figure 3.3) exhibits an increasing divergence as the scaling multiplier increases. This behavior highlights that the Diamond and Gyroid geometries yield superior hydraulic performance, providing more effective pressure drop management at higher operational flow scales.



In all considered scenarios, whether porosity is explicitly included or neglected in the formulations, the performance trends remain consistent. This consistency confirms the robustness of the computational model and the validity of the underlying engineering assumptions.

Having elucidated the thermal-hydraulic behavior of the different topologies, it is crucial to evaluate their performance as a function of the unit cell size.

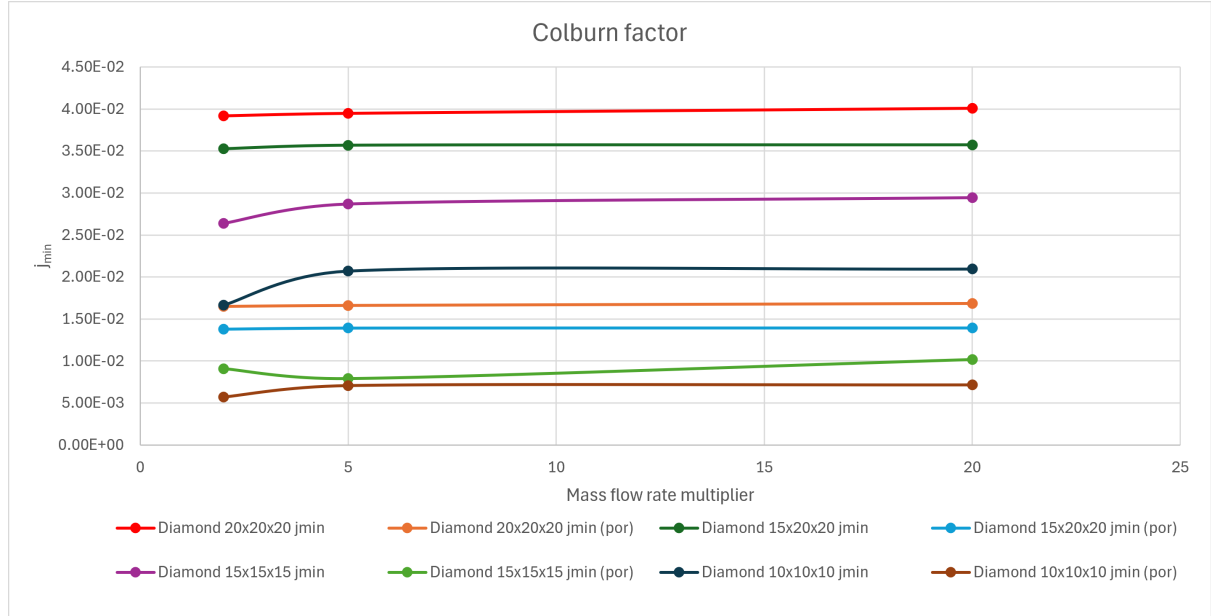


Figure 3.4: Colburn j -factor evaluation across different cell sizes, comparing formulations with and without porosity inclusion.

Intuitively, one might expect that decreasing the cell size—thereby increasing the specific surface area and the total heat transfer area—would enhance the convective heat transfer and, consequently, the Colburn factor. However, the observed trend differs. This counter-intuitive behavior can be analytically explained by expanding the j -factor definition into its primary operational variables³:

$$j = \frac{\dot{Q}}{A_{wet} \cdot \Delta T} \frac{A_{in}}{\dot{m} \cdot c_p^{1/3}} \left(\frac{\mu}{k} \right)^{2/3} \quad (3.6)$$

Equation 3.6 demonstrates that for a given thermal duty and mass flow rate, an increase in the wetted surface area inherently reduces the j -factor.

³In this context, “primary variables” refer to the direct boundary conditions and geometric inputs, with the exception of the heat transfer rate $\dot{Q}$, which is directly computed via the STAR-CCM+ simulations.

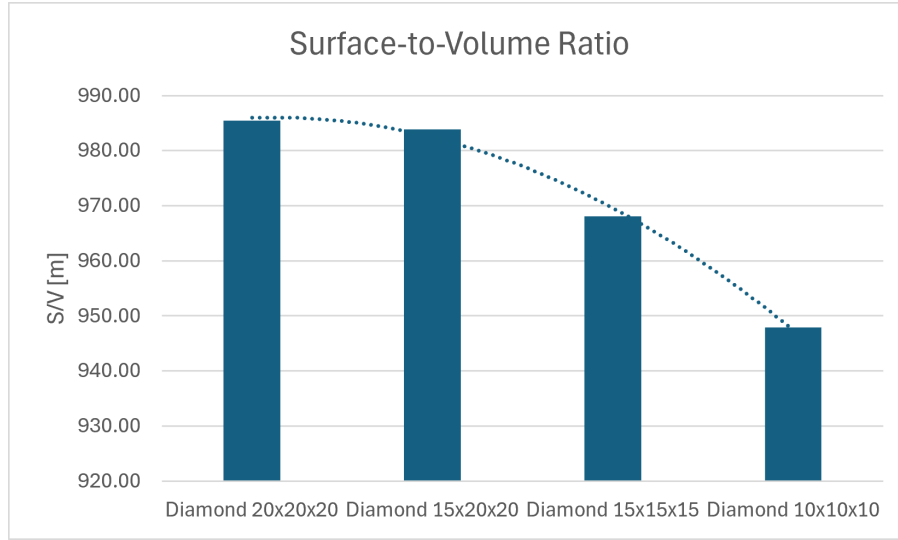


Figure 3.5: Evolution of the specific surface area across different unit cell sizes.

Since the specific surface-to-volume ratio strictly decreases as the cell size decreases (Fig. 3.5), the Colburn factor experiences a mathematical penalization. Furthermore, the fluid dynamic penalty induced by the restriction of the fluid channels becomes profoundly evident when analyzing the goodness factor, which explicitly accounts for pressure drops (Fig. 3.6).

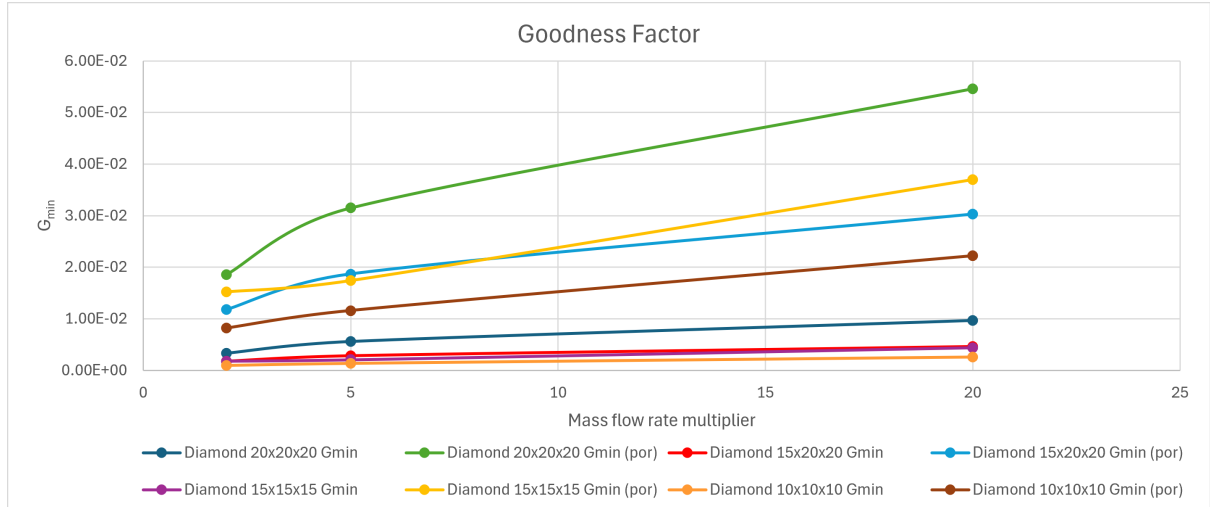


Figure 3.6: Goodness factor comparison for different unit cell sizes, illustrating the severe impact of pressure drops in smaller geometries.

An additional aspect of interest investigated in this study is morphological anisotropy, specifically anisotropic cell scaling. A configuration deformed along the flow direction was tested, maintaining the exact transverse cross-section of the 20 mm cubic cell while reducing the longitudinal pitch to increase the number of cells in the flow direction ($15 \times 20 \times 20$ mm unit cell).

Although this anisotropic grading yielded satisfactory performance across both evaluation metrics, the isotropic $20 \times 20 \times 20$ mm³ cubic cell proved superior. Moreover, considering that the $15 \times 20 \times 20$ mm³ topology required a larger volume of solid resin, thereby increasing manufacturing costs and overall weight, it was ultimately discarded.



During the initial testing phases at the nominal design flow rate, the computed thermal efficiencies asymptotically approached 100% across all configurations. This asymptotic behavior flattened the performance curves, making it difficult to effectively benchmark the topologies. To overcome this limitation and artificially force a higher flow velocity within the cold stream, which inherently exhibited lower velocities than the hot stream, the SplitP architecture was investigated. This topology was specifically selected because it exhibits distinct directional dependencies and non-self-complementary fluid domains, resulting in drastically different hydraulic impedances between the two channels. However, this strategy proved counterproductive during the scaled-up mass flow simulations. Furthermore, the SplitP geometry presented severe AM challenges, rendering the heat exchanger unfeasible to print and leading to its definitive abandonment.

Given the absence of stringent design constraints on pressure drops, it was methodologically valuable to explore highly restrictive geometries to assess potential heat transfer maximization. Nevertheless, CFD simulations conclusively demonstrated that the isotropic Gyroid and Diamond topologies, equipped with a 20 mm cell size, remain the optimal engineering solutions for the problem at hand.

In light of the obtained results, the primary design objectives, and the underlying assumptions, the theoretical optimal configuration would have been a Gyroid TPMS featuring a $20 \times 20 \times 20 \text{ mm}^3$ unit cell and a wall thickness of 1 mm. However, this ideal geometry proved unfeasible due to time constraints and technological limitations encountered during the 3D printing process. Consequently, to ensure successful manufacturability, the final design selection was necessarily redirected toward a Diamond topology with a $15 \times 15 \times 15 \text{ mm}^3$ unit cell and a conservatively increased wall thickness of 1.8 mm.

Future investigations should systematically evaluate the scaling of the unit cell dimensions to identify the specific geometric thresholds at which larger periodic lengths optimize the overall thermo-hydraulic performance. Because the thermal effectiveness and hydrodynamic resistance of TPMS heat exchangers are highly sensitive to variations in pore diameter and periodic length.



4 CAD Modelling

To generate the complete HX assembly comprising the outer shell, manifolds, and internal core suitable for numerical simulation, a synergistic workflow involving two distinct computational design tools is essential. Specifically, SolidWorks is employed for the parametric modeling of the solid casing, the manifolds, and the macroscopic fluid domains. Conversely, nTopology is utilized to generate the complex Triply Periodic Minimal Surface structures (implicit modeling) and to perform the boolean extraction of the micro-scale fluid domains required for the CFD simulations in Simcenter STAR-CCM+.

4.1 SolidWorks Modeling: Shell, Manifolds, and Fluid Domain

The initial phase of the CAD workflow is conducted within the SolidWorks environment, focusing on the design of the macroscopic structural components and the definition of the global fluid boundaries.

4.1.1 Shell and Manifolds Design

The external shell is explicitly modeled to embed a few millimeters of the internal TPMS core, ensuring structural integrity and robust flow containment. Although optimizing the manifold geometry to evaluate variations in global performance represents a valid avenue for future investigations, it is not the primary focus of this preliminary study. Instead, a functional and simplified approach was adopted for the fluid distribution.

The manifolds were designed utilizing solid core extrusions that penetrate the TPMS matrix to inject the working fluids deeper into the heat exchanger core. To ensure strict fluidic isolation and prevent cross-contamination between the two distinct fluid domains inherently generated by the sheet-based TPMS architecture, precise boolean operations were executed within the nTop environment. For instance, at the hot fluid inlet manifold, all geometric intersections between the penetrating core and the cold fluid domain were explicitly sealed. This methodology guarantees that the incoming fluid is routed exclusively into its designated channels without mixing, while successfully delivering the flow into the active convective heat transfer region.

Furthermore, to facilitate the physical integration of the final additively manufactured prototype into a hydraulic test bench, standardized threaded connections are directly incorporated into the manifold design, ensuring secure and leak-tight fluidic couplings.

4.1.2 Macroscopic Fluid Envelope Definition

A critical preparatory step for the subsequent topology generation and CFD analysis is the definition of the macroscopic fluid envelope. Unlike standard CFD workflows where the traditional CAD software directly extracts the final fluid domain, in this phase SolidWorks is utilized solely to model the bulk internal cavity defined by the outer shell and the manifolds.

Through initial boolean operations, a rigorous, watertight, and topologically clean 3D solid representing the global void volume is generated. This bulk envelope does not



represent the final computational domain; rather, it serves as the essential boundary input for the subsequent modeling phase.

4.2 nTopology: TPMS Lattice Generation and Fluid Domain Extraction

In the second phase of the geometrical setup, the computational design software nTop is employed to generate the complex Triply Periodic Minimal Surface architectures and extract the final flow regions. Initially, the macroscopic boundary components generated in SolidWorks are imported into the nTop environment as independent solid bodies. Because TPMS geometries are defined fundamentally by implicit mathematical functions, these imported CAD models are subsequently converted into implicit representations using the *Implicit Body from CAD Body* operator. Finally, these implicit boundaries are utilized to spatially constrain and intersect the mathematical TPMS lattice, extracting the exact fluid domains required for the downstream CFD simulations in Simcenter STAR-CCM+.

4.2.1 TPMS Lattice Definition

To mathematically generate the TPMS matrix, several foundational design variables must first be defined:

- **Topology:** Diamond;
- **Unit cell dimensions:** $15 \times 15 \times 15 \text{ mm}^3$;
- **Wall thickness:** 1.8 mm;
- **Reference Frame:** Establishing the local coordinate system, comprising the spatial origin and directional vectors, is essential for generating the trigonometric TPMS architecture. The origin ensures a symmetric unit cell distribution, while the directional axes allow the internal flow channels to be rotated independently of the macroscopic fluid envelope.

These input parameters dictate the creation of the fundamental TPMS unit cell, which is subsequently patterned and expanded across a three-dimensional grid. The final spatial extent of this periodic lattice is strictly constrained and bounded by the macroscopic fluid envelope imported from SolidWorks, ensuring a perfect geometrical match for the subsequent boolean fluid extraction.

4.2.2 Heat Exchanger Assembly and Fluid Domain Extraction

Following the mathematical generation of the TPMS lattice, the complete HX must be assembled. A critical phase in this process is ensuring that the fluids are properly routed from the manifolds into their respective channels without cross-contamination. Because a sheet-based TPMS inherently divides the spatial volume into two distinct, non-intersecting fluid labyrinths, the manifold inlets must be geometrically isolated. Solid core extrusions extending from the manifolds penetrate a few millimeters into the lattice matrix. Boolean subtractions and intersections are then applied as described before.

Once the structural assembly of the heat exchanger is finalized, further boolean operations are required to extract the exact computational domains for the numerical simulations.

Specifically, the solid TPMS matrix is subtracted from the macroscopic bounding envelope to explicitly generate the internal fluid cavities. This procedure yields three distinct, watertight 3D bodies: the solid HX core and the two separate fluid domains.

Finally, the continuous volumetric regions defined by the implicit mathematical functions must be converted into a discrete format compatible with the FVM employed by the downstream CFD solver. Because the FVM framework transforms continuous partial differential equations governing fluid motion into a system of discrete algebraic equations solved over non-overlapping control volumes, the boundary surfaces of the domain must be explicitly tessellated.

Within the nTopology environment, this initial discretization is executed utilizing the *Mesh from Implicit Body* operator (Fig. 4.1). In this specific methodological workflow, this function is employed to tessellate strictly the exact implicit boundaries of the solid TPMS matrix into a high-resolution surface mesh, typically composed of triangular elements. The resulting discrete surface body, representing exclusively the solid shell, is subsequently exported in a format file (.stl).



Figure 4.1: Implementation of the *Mesh from Implicit Body* block in nTopology to generate the discrete surface mesh of the solid TPMS matrix prior to export.

Rather than pre-extracting the complex fluid labyrinths during the nTopology phase, this approach imports the discrete solid TPMS mesh alongside the macroscopic fluid envelope (the bulk volume) directly into Simcenter STAR-CCM+. The actual fluid domains are then generated internally within the CFD software by executing a boolean subtraction of the solid TPMS matrix from the bulk fluid block. These imported geometries act as the foundational boundary inputs, ensuring a perfectly conformal fluid-solid interface when the final volumetric computational grid is generated.



5 CFD Simulation and Solution Verification

Performing a rigorous CFD analysis requires discretizing the continuous fluid domain into a high-quality computational grid, a step that fundamentally governs the accuracy and stability of the numerical solver. However, obtaining a converged result on a single, computationally convenient mesh is insufficient for engineering certification. It is strictly necessary to perform Solution Verification to quantify the discretization error and mathematically estimate how far the computed results deviate from the exact analytical continuum solution at zero grid spacing.

5.1 Mesh Generation and Simulation Setup

Once the tessellated solid geometry is imported from nTopology, internal Boolean operations within Simcenter STAR-CCM+ are executed to generate the distinct fluid domains by subtracting the TPMS structure from the macroscopic fluid envelope. The resulting geometric parts are subsequently mapped into discrete computational regions, each assigned to a corresponding Physics Continuum that dictates the governing mathematical models and thermodynamic properties.

5.1.1 Physics Continua

In STAR-CCM+, a Physics Continuum encapsulates the system of governing partial differential equations, physical state models, and material properties applied to a given computational region. To resolve the Conjugate Heat Transfer problem between the working fluids and the conductive solid structure, three distinct continua are instantiated: two fluid continua (*Cold Air* and *Hot Air*) and one solid continuum (*Resin*).

Although both flow channels transport the same physical medium (air) and employ identical governing equations, separating them into two independent fluid continua was strategically adopted to initialize each stream with distinct thermodynamic states (temperatures and velocity). Supplying dedicated initial conditions close to their respective boundary inlet values substantially reduces initial unphysical gradients across the solid interface, enhancing numerical stability and accelerating the convergence rate toward the steady-state solution.

The physical models and numerical formulations selected for each continuum are detailed below:

- **Fluid Continua (Cold Air and Hot Air):**

- *Three-Dimensional and Steady:* Resolves the time-averaged conservation equations in a 3D domain under stationary conditions.
- *Gas and Ideal Gas Law:* Treats air as a compressible ideal gas, evaluating local density variations via the equation of state:

$$\rho = \frac{pV}{RT} \quad (5.1)$$



- *Laminar Flow*: Solves the laminar Navier–Stokes and continuity equations directly, justified by the low-Reynolds-number regime characteristic of the nominal TPMS core channels⁴.
- *Segregated Flow*: Solves the momentum and pressure equations sequentially using a co-located pressure-velocity coupling scheme (SIMPLE-type).
- *Segregated Fluid Temperature*: Resolves the convective-diffusive thermal energy transport equation independently from momentum, updating convective fluxes at each iteration.
- *Gradients*: Computes spatial gradients of flow variables (velocity, pressure, and temperature) across cell faces using a cell-based reconstruction scheme, essential for evaluating diffusive fluxes and secondary gradients.
- **Solid Continuum (Resin)**:
 - *Three-Dimensional, Steady, and Solid*: Formulates thermal conduction across the physical matrix of the 3D-printed photopolymer resin.
 - *Constant Density*: Employs a fixed density (ρ_{resin}) across the entire operational temperature range.
 - *Segregated Solid Energy*: Discretizes Fourier’s law of thermal conduction:

$$\nabla \cdot (k_{\text{resin}} \nabla T) = 0 \quad (5.2)$$

disabling momentum transport equations within the solid matrix.

- *Gradients*: Reconstructs temperature gradients at cell faces and boundary interfaces to calculate conductive heat fluxes.

Additionally, the solver adds in the all three continua the *Solution Interpolation* that works as numerical model used to interpolate solution data from one mesh to another when meshes have no perfect matching faces/cells at the interface.

5.1.2 Computational Regions and Boundary Conditions

The discretized geometry is partitioned into three independent regions: the cold fluid channel, the hot fluid channel, and the solid TPMS core. The boundary conditions applied across the external and internal interfaces of these regions are defined as follows:

- **Mass Flow Inlet**: Imposed at the inlet boundaries of both the hot and cold fluid regions. It enforces a specified uniform mass flow rate and bulk inlet temperature, determining the inlet velocity profile based on local cross-sectional area and fluid density.
- **Pressure Outlet**: Prescribed at the downstream outlets of both flow channels, setting a fixed static relative pressure ($p = 0$ Pa).
- **External Wall**: Represents the outer casing of the heat exchanger exposed to ambient conditions, modeled hydrodynamically using a standard no-slip kinematic condition ($\vec{u} = \vec{0}$). Thermally, this boundary is evaluated as a convective heat transfer condition (Robin boundary condition), which imposes a fixed ambient bulk temperature (T_{∞}) alongside an external heat transfer coefficient (h_{ext}) to account

⁴The laminar model is justified as these simulations are conducted exclusively at the nominal flow rate, ensuring a purely laminar regime.



for natural convection and a thermal resistance to simulate the thickness of the outer shell. Since we know the external temperature during the tests, the thickness of the casing and the thermal conductivity of the resin by design, the only missing parameter is h_{ext} . Starting from the convective heat transfer relation,

$$h_{\text{ext}} = \frac{\dot{Q}_{\text{loss}}}{A_{\text{ext}} (T_{\text{wall,ext}} - T_{\text{amb}})}, \quad (5.3)$$

it can be observed that the unknown quantity is the external shell-wall temperature. Since the shell is not explicitly included in the numerical model, this temperature cannot be directly obtained from the iterative simulation. Therefore it was applied the following procedure for a first approximated value that should be checked. The last 100 values of the four experimentally measured temperatures from Test 2 were considered, namely the inlet and outlet temperatures of both the hot and cold streams. For each stream, the corresponding mean fluid temperature was calculated as

$$T_{\text{mean}} = \frac{T_{\text{in}} + T_{\text{out}}}{2}. \quad (5.4)$$

The temperature of the internal shell-wall surface was assumed equal to the mass-flow-weighted average of the mean temperatures of the two streams:

$$T_{\text{wall,int}} = \frac{\dot{m}_h T_{\text{mean,h}} + \dot{m}_c T_{\text{mean,c}}}{\dot{m}_h + \dot{m}_c}. \quad (5.5)$$

Actually this temperature is roughly representative of the internal shell-wall temperature because we should consider a logarithmic mean temperature difference for the bulk average, the internal convective heat transfer coefficient and mainly the temperature difference and variation between the fluids. Subsequently, knowing the thermal resistance of the shell wall, the external wall temperature was determined by applying Fourier's law for a plane wall:

$$\dot{Q}_{\text{loss}} = \frac{k_{\text{wall}} A_{\text{wall}}}{\delta_{\text{wall}}} (T_{\text{wall,int}} - T_{\text{wall,ext}}), \quad (5.6)$$

where the total heat-transfer area A_{wall} , was calculated using the mean-plane approximation. By rearranging the previous equation, the unknown external shell-wall temperature was obtained as:

$$T_{\text{wall,ext}} = T_{\text{wall,int}} - \frac{\dot{Q} \delta_{\text{wall}}}{k_{\text{wall}} A_{\text{wall}}}. \quad (5.7)$$

It was then substituted into the inverse convective heat transfer relation to determine the external heat transfer coefficient h_{ext} with a final value of $18 \text{ W}/(\text{m}^2 \text{ K})$.

- **Fluid–Solid Interface (Conjugate Heat Transfer Interface):** The wetted, interpenetrating internal surfaces separating the fluid labyrinths from the solid TPMS wall. Rather than a simple boundary, this is configured as an in-place conformal interface coupling the adjacent fluid and solid regions. It enforces a no-slip hydrodynamic condition alongside continuous heat flux and temperature across the boundary:

$$T_{\text{fluid}} = T_{\text{solid}}, \quad -k_{\text{fluid}} \left. \frac{\partial T_{\text{fluid}}}{\partial n} \right|_{\text{int}} = -k_{\text{solid}} \left. \frac{\partial T_{\text{solid}}}{\partial n} \right|_{\text{int}} \quad (5.8)$$

enabling conjugate convective–conductive heat transfer between the hot stream, the conductive resin core, and the cold stream.



5.1.3 Computational Mesh Generation

The spatial discretization of the computational domain for complex topologies, such as TPMS, necessitates a hybrid meshing strategy to reconcile geometric conformity with numerical rigor. Within the bulk regions, grid generation leverages an unstructured topology based on polyhedral elements. These elements exhibit superior morphological flexibility, enabling seamless adaptation to the tortuous channels of the TPMS matrix while substantially reducing the overall cell count compared to standard isotropic tetrahedral formulations.

However, deploying isotropic polyhedral elements directly adjacent to solid interfaces is numerically deficient. Resolving the near-wall boundary layer using isotropic cells would require an unfeasible number of elements and introduce severe artificial numerical diffusion, primarily because the cell faces fail to align with the wall-parallel streamlines. To mitigate this issue and accurately resolve the steep near-wall velocity and temperature gradients, implementing a structured prism layer stack is mandatory. This near-wall sub-grid consists of prismatic cells aligned parallel to the bounding surface with an high aspect ratio, effectively suppressing wall-normal numerical diffusion.

The architecture of this prismatic stack is defined by four interdependent parameters: the first cell height (Δy), total thickness (H), layer count (N_{layers}), and stretching factor (r). Crucially, while these parameters are conventionally designed to satisfy turbulent wall-resolution criteria ($y^+ \approx 1$), this structured refinement remains vital for the low-Reynolds-number laminar regime investigated herein. Even in the absence of a turbulent viscous sublayer, the prism layer is imperative to capture the intense thermal gradients at the fluid–solid interface and to accurately predict the complex, eccentric helical flows that dictate TPMS convection. Consequently, this hybrid configuration captures wall shear stresses accurately, prevents the underestimation of pressure drop, and ensures high numerical fidelity in the CHT calculations.

To optimize computational resources, a surface control was implemented to disable prism layers on the solid side of the interface, reserving structured boundary-layer refinement exclusively for the fluid flow because heat transport within the resin matrix is governed purely by steady Fourier conduction. However, also inside the solid region, there is a temperature gradient and therefore possible numerical diffusion; thus, the use of some prism layer in this region should be investigated in the future.

The comprehensive baseline parameters and mesh controls are summarized in Table 5.1.



Table 5.1: Mesh parameters used for the TPMS simulations

Parameter	Value
<i>Meshers</i>	
Surface Remesher	
Polyhedral Mesher	
Prism Layer Mesher (Distribution Mode: Stretching)	
<i>Default Controls</i>	
Base Size	1.0 mm
Target Surface Size	100% (Base Size)
Minimum Surface Size	10% (Base Size)
Surface Curvature	36 points/circle
Surface Growth Rate	1.2
Volume Growth Rate	1.2
<i>Custom Controls – Volume Control → Cold & Hot fluid</i>	
Polyedral Mesher	50 % (Base Size)
<i>Prism Layer</i>	
Prism Layer Total Thickness	0.35 mm (Absolute)
Number of Prism Layers	4
Prism Layer Stretching Factor	1.15
Gap Fill Percentage	25%
Minimum Thickness Percentage	30%
<i>Custom Controls – Surface Control → Solid</i>	
Prism Layer	Disable

5.2 Solution Verification and Grid Independence

In computational fluid dynamics, establishing numerical credibility requires a rigorous mathematical assessment to quantify discretization errors (δ_{num}) independently of physical assumptions, as mandated by the ASME V&V 20-2009 standard. Obtaining converged iterations on a single computational grid cannot guarantee that the discretized solution accurately mirrors the continuous governing equations. Therefore, prior to physical validation against experimental data, a formal Solution Verification is conducted to assess grid independence and quantify the numerical uncertainty (u_{num}) arising from spatial discretization.



5.2.1 Methodological Framework: Richardson Extrapolation and GCI

To estimate the spatial discretization error, the Grid Convergence Index (GCI) methodology, rooted in generalized Richardson Extrapolation, is implemented.

Following the ASME V&V 20 guidelines, a systematic five-step procedure is executed:

Step 1: Definition of Representative Grid Size

Because the computational domain features an unstructured hybrid grid (polyhedral core cells coupled with near-wall prismatic layers), the representative cell size h cannot be computed from Cartesian coordinates. It is instead evaluated via the volumetric mean of all N elements within the domain:

$$h = \left[\frac{1}{N} \sum_{i=1}^N \Delta V_i \right]^{1/3} \quad (5.9)$$

Step 2: Systematic Grid Refinement

Three geometrically similar computational grids are generated in Simcenter STAR-CCM+: a Coarse grid (h_3), a Medium grid (h_2), and a Fine grid (h_1). The grid refinement factor r is defined as:

$$r_{21} = \frac{h_2}{h_1}, \quad r_{32} = \frac{h_3}{h_2} \quad (5.10)$$

To ensure sufficient separation between the physical signal and numerical truncation noise, the refinement ratio is maintained near the recommended value ($r \approx 1.30$), scaling both the bulk polyhedral base size and the prism layer dimensions proportionally to preserve cell aspect ratios.

As a first step, a refinement ratio of $r = 1.5$ was assumed and applied to both meshes. Starting from the medium grid, with a base size of 1 mm, the base sizes of the fine and coarse grids were obtained as 0.67 mm and 1.5 mm, respectively.

To maintain consistency between the three grids, all the Polyhedral Mesher parameters were defined as percentages of the base size. For the Prism Layer Mesher, the total prism layer thickness and the stretching factor were kept constant, as reported in Table 5.1. This ensured that the same region of the velocity and temperature profiles was resolved in all three grids, while allowing the prism-layer mesh to become progressively finer. In fact, the number of prism layers was varied according to the refinement ratio r , resulting in 7, 10, and 15 layers for the coarse, medium, and fine grids, respectively.

As expected for a non-uniform mesh, the refinement ratio imposed on the base size resulted in calculated refinement ratios based on the characteristic mesh size that were lower than the nominal value. In the present case, these values were even below the minimum recommended value of $r = 1.3$. Therefore, in order to obtain usable refinement ratios, the base size of the medium grid was changed to 0.8 mm, and the Surface Curvature and Surface Proximity parameters were adjusted accordingly. The resulting mesh parameters are reported in Table 5.2.



Steps 3 & 4: Order of Convergence and Extrapolated Values

Rather than calculating the apparent order of convergence p directly from the discrete solution data, a formal theoretical order of $p = 2$ is strictly assumed. This prescribed value aligns directly with the mathematical accuracy of the numerical discretization schemes employed in the simulation model, specifically the second-order upwind scheme utilized for the reconstruction of energy and momentum fluxes, alongside the second-order spatial scheme applied for the pressure term discretization.

By enforcing this theoretical order of convergence, the extrapolated continuum solution f_{ext}^{21} at zero grid spacing ($h \rightarrow 0$) is evaluated as:

$$f_{\text{ext}}^{21} = \frac{r_{21}^p f_1 - f_2}{r_{21}^p - 1} \quad (5.11)$$

where f_1 and f_2 represent the discrete solutions obtained on the fine and medium grids, respectively, and r_{21} is the grid refinement ratio.

Step 5: Evaluation of Numerical Uncertainty via GCI

The relative approximate error between the fine and medium solutions is given by:

$$e_a^{21} = \left| \frac{f_1 - f_2}{f_1} \right| \quad (5.12)$$

The fine-grid convergence index ($\text{GCI}_{\text{fine}}^{21}$), representing the expanded numerical uncertainty within an approximate 95% confidence interval, is calculated as:

$$\text{GCI}_{\text{fine}}^{21} = \frac{F_s \cdot e_a^{21}}{r_{21}^p - 1} \quad (5.13)$$

In accordance with the ASME standard, a Factor of Safety $F_s = 3.0$ is adopted, to be more conservative. Finally, the standard numerical uncertainty (u_{num}) representing a 1σ interval is extracted by accounting for the shifted Gaussian distribution centered on f_{ext}^{21} :

$$u_{\text{num}} = \frac{\text{GCI}_{\text{fine}}^{21}}{1.15} \quad (5.14)$$

5.2.2 Grid Convergence Results and Key Target Metrics

The verification procedure is conducted on five critical performance indicators of the TPMS heat exchanger: the total core pressure drop (Δp) across hot and cold channels, the logarithmic mean temperature difference (ΔT_{lm}) and the Heat transfer rate ($\dot{Q}$) for hot and cold side. Although in the experimental set-up there are not pressure transducers and therefore no direct possible comparison with simulations, it is still an important parameter for the analysis of the fluid-dynamic numerical stability.

The Grid Convergence Index analysis was carried out on a scaled-down computational volume of $15 \times 15 \times 90 \text{ mm}^3$, owing to the high computational overhead of resolving intricate TPMS geometries. This reduced domain substantially accelerated iteration times, enabling a thorough exploration of diverse meshing strategies and refinement ratios while preserving the geometric periodicity and representative flow physics of the core, imposing periodic boundary conditions on the four later faces.



The results are shown in the Table 5.2.

Table 5.2: Discretization error evaluation and GCI calculation for the TPMS core under nominal flow conditions (assuming a theoretical order of convergence $p = 2$).

Target Metric	Coarse (f_3)	Medium (f_2)	Fine (f_1)	GCI_{fine}^{21}	u_{num}
<i>Global Grid Parameters</i>					
Number of Cells (N)	1283940	4766314	10708067	—	—
Refinement Ratio (r)	—	1.27 (r_{32})	1.55 (r_{21})		
<i>Thermal Field Performance</i>					
ΔT_{lm} [K]	3.58592	3.74486	3.73874	0.687 %	0.597 %
Q_{hot} [W]	0.40660	0.40760	0.40790	0.312 %	0.271 %
Q_{cold} [W]	-0.40656	-0.40757	-0.40792	0.361 %	0.314 %
<i>Hydraulic Field Performance</i>					
Δp_{hot} [Pa]	29.04438	28.35302	28.44939	1.827 %	1.588 %
Δp_{cold} [Pa]	2.66466	2.70365	2.70182	0.563 %	0.490 %

By enforcing the theoretical order of convergence ($p = 2$), the calculated Grid Convergence Index mathematically falls below 1% for the majority of the evaluated thermal and hydraulic metrics. However, this prescribed formulation may obscure localized discretization errors; in highly complex physical domains, geometric singularities and high local gradients can cause the true observed order to drop significantly below the theoretical expectation. The hot-side pressure drop (Δp_{hot}) exposes this underlying sensitivity, exhibiting a worst-case GCI of approximately 1.83% alongside minor convergence oscillations. Given the inherent uncertainties introduced by bypassing the directly calculated apparent order, conservative V&V practices mandate adopting this maximum discrepancy to bound the standard numerical uncertainty (u_{num}) in the subsequent validation phase. Consequently, while the current fine grid provides a nominally acceptable baseline, its asymptotic independence is regarded with caution. Future mesh generation strategies must introduce targeted localized volumetric controls to resolve the pressure field anomalies and strictly verify the validity of the assumed convergence behavior.



6 Additive Manufacturing and Experimental Testing

The complex, non-intersecting, and interpenetrating fluid labyrinths defined by TPMS cannot be fabricated using traditional subtractive manufacturing techniques such as CNC milling or casting. Additive Manufacturing serves as the critical enabler that translates these mathematically defined architectures into functional physical hardware. However, the physical realization of micro-architected geometries introduces significant practical constraints, defect mechanisms, and experimental challenges. This chapter details the manufacturing methodology adopted for the Diamond TPMS heat exchanger, characterizes the physical implications of resin-based 3D printing, and describes the experimental test facility designed to validate the numerical CFD model under controlled operating conditions.

6.1 Additive Manufacturing Strategy and Fabrication Constraints

6.1.1 Process Selection: SLA vs. Powder Bed Fusion

While metal-based additive manufacturing processes, such as SLM and L-PBF, are conventionally employed for high temperature operational heat exchangers due to their elevated thermal conductivity (e.g., AlSi10Mg or pure copper), photopolymer resin stereolithography (SLA/DLP) was selected for this investigation. Vat photopolymerization provides exceptional geometric fidelity, high surface smoothness, and precise feature resolution at sub-millimeter scales, making it an ideal platform for hydro-thermal aerodynamic visualization and low-temperature experimental benchmarking.

6.1.2 Design for Additive Manufacturing (DfAM) and Printing Constraints

The transition from idealized CAD representations to physical artifacts requires strict adherence to DfAM rules:

- **Self-Supporting Overhangs:** TPMS structures naturally exhibit continuous local curvature. However, as identified in preliminary topological investigations, aggressive geometries such as SplitP present extreme downskin overhang angles that fail to self-support during layer-by-layer deposition, leading to structural collapse without sacrificial internal supports. Conversely, the chosen isotropic Diamond topology (15mm unit cell) maintains local surface inclination angles that safely eliminate the need for internal support structures, ensuring open, unobstructed fluid channels.
- **Wall Thickness Selection:** While a minimal wall thickness of 1.0 mm was theoretically optimal to reduce solid conductive resistance, manufacturing constraints mandated an increase to 1.8 mm. This structural reinforcement prevents membrane deflection induced by peeling forces during vat photopolymerization and ensures di-

mensional repeatability across the core.

- **Uncured Resin Evacuation:** Because sheet-based TPMS divides the domain into two completely isolated labyrinths, dedicated drainage pathways and post-processing flushing protocols were integrated into the manifold design to ensure total evacuation of uncured liquid resin prior to final UV curing.

6.1.3 Impact of Manufacturing Defects on Thermo-Hydraulics

Additively manufactured surfaces deviate systematically from theoretical geometries. In powder-based systems, stair-stepping effects and partially sintered particles on downskin surfaces introduce severe roughness, which can increase experimental pressure drops by up to 69% relative to smooth CAD predictions. Although photopolymer resin printing drastically mitigates granular roughness, localized dimensional tolerances and layer interfaces introduce micro-scale variations that perturb near-wall laminar flow profiles and alter boundary layer development.

6.2 Experimental Facility and Instrumentation

To validate the numerical conjugate heat transfer model, an experimental open-loop wind tunnel test facility was assembled. The apparatus is explicitly configured to isolate the test section from external thermal interactions, thereby ensuring an accurate evaluation of the global energy balances across the heat exchanger core.

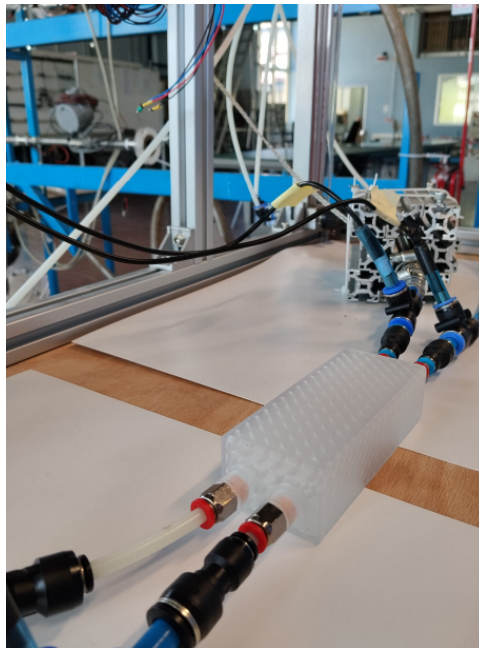


Figure 6.1: Photograph of the assembled experimental thermal-hydraulic test facility, illustrating the airflow channel and the instrumentation layout.

6.2.1 Flow Loop and Fluid Delivery

The test facility comprises two decoupled flow circuits operating with air as the working fluid:

- **Fluid Delivery and Control:** Air is directed through a dedicated flow channel equipped with an upstream flow straightener to suppress macroscopic swirl and ensure a uniform velocity profile at the test section inlet. The mass flow rate is precisely modulated via control valves, and its variations are continuously monitored using an in-line flow meter.

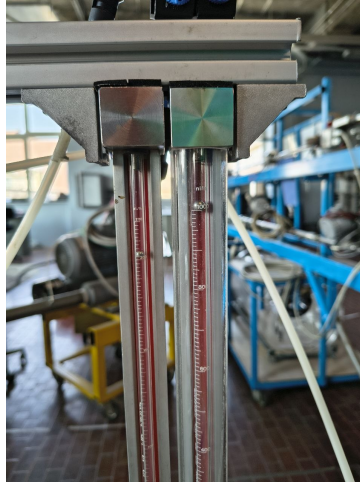


Figure 6.2: In-line flow meter configuration utilized for the continuous monitoring and modulation of the inlet air mass flow rate.

- **Thermal Conditioning:** The fluid stream is thermally conditioned upstream utilizing an electrically controlled heating unit, which is governed by a precision temperature controller. To accurately evaluate the thermal performance, the fluid temperatures are continuously monitored using calibrated thermocouples, ensuring that experimental data is collected strictly under stabilized, steady-state thermal conditions.

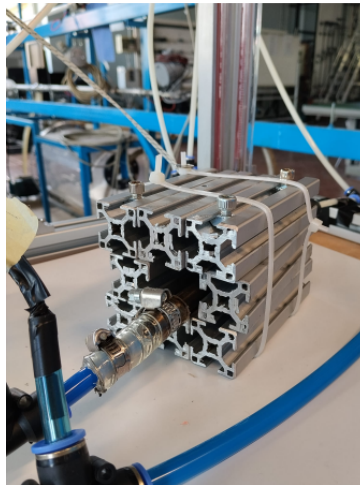


Figure 6.3: Electrically controlled heating unit and precision temperature controller employed for the steady-state thermal conditioning of the working fluid.



6.2.2 Sensor Integration and Data Acquisition

To accurately capture the thermal boundary conditions across the test section, a custom digital data acquisition (DAQ) system was implemented utilizing an Arduino microcontroller platform. The raw data acquired by the microcontroller are processed through a dedicated Python script and continuously logged into a *.csv* file for subsequent post-processing. For temperature measurements, calibrated thermocouples are installed at the inlet and outlet manifolds of both fluid streams to monitor and record the bulk fluid temperatures (T_{in} and T_{out}). This continuous data logging is critical to ensure that experimental measurements are extracted strictly after a stable, steady-state thermal condition has been achieved throughout the testing loop.

7 Post-Processing and Model Validation

This chapter presents the quantitative comparison between the full-scale Conjugate Heat Transfer (CHT) numerical simulations and the experimental measurements collected on the physical heat exchanger prototype. In accordance with the ASME V&V 20-2009 standard methodology, the objective is to isolate and quantify the modeling error (δ_{model}) by assessing both numerical and experimental uncertainties.

7.1 Experimental Setup and ASME V&V 20 Validation Methodology

To evaluate the predictive capability of the computational model across the operational envelope, five distinct experimental test runs were executed. The operating conditions were specified by matching the mass flow rates ($\dot{m}$) of both the hot and cold air streams across the laboratory test facility and the 3D computational domain.

Model validation does not rely on subjective graphical alignment, but rather on a formal mathematical comparison between the comparison error (E) and the validation uncertainty (u_{val}). The raw comparison error between the Simcenter STAR-CCM+ simulation (S) and the experimental measurement (D) is defined as:

$$E = S - D \quad (7.1)$$

Because both the numerical prediction and the physical experiment are subject to uncertainties, the validation uncertainty (u_{val}) is formulated by combining the independent sources of error in quadrature:

$$u_{\text{val}} = \sqrt{u_{\text{num}}^2 + u_{\text{input}}^2 + u_D^2} \quad (7.2)$$

where the three primary sources of uncertainty are formally defined as follows:

- **Experimental Measurement Uncertainty (u_D):** Arising from the physical instrumentation, specifically driven by the temperature transducers with a standard measurement accuracy of $\pm 1.0^\circ\text{C}$.
- **Numerical Discretization Uncertainty (u_{num}):** Extracted via the Grid Convergence Index (GCI) procedure to quantify the error induced by spatial discretization.
- **Input Uncertainty (u_{input}):** Derived from variations and approximations in boundary conditions and thermophysical parameters, such as the external convective heat transfer coefficient (h_{ext}) representing ambient heat losses.

The true modeling error (δ_{model}), arising from physical idealizations such as the laminar assumption or smooth wall formulation, is then formally bounded within the confidence interval:

$$\delta_{\text{model}} \in [E - u_{\text{val}}, E + u_{\text{val}}] \quad (7.3)$$



7.2 Results and Validation Analysis

The quantitative assessment of the CHT numerical model against laboratory experimental measurements was conducted across four distinct test regimes. The numerical predictions (S) and experimental measurements (D) for the hot fluid stream temperature drop (ΔT_{hot}), cold fluid stream temperature rise (ΔT_{cold}), and parasitic thermal power losses ($\dot{Q}_{\text{loss}}$) are summarized in Table 7.1, alongside the resulting comparison errors ($E = S - D$).

Table 7.1: Quantitative comparison between CHT simulation predictions (S) and experimental measurements (D) across the four tested flow rate configurations.

Test ID	Target Metric	Simulation (S)	Experiment (D)	Comparison Error (E)
Test 1	ΔT_{hot} [K]	11.93	15.20	-3.27
	ΔT_{cold} [K]	18.99	7.92	+11.07
	$\dot{Q}_{\text{loss}}$ [W]	3.28	5.45	-2.17
Test 2	ΔT_{hot} [K]	12.24	14.54	-2.30
	ΔT_{cold} [K]	18.88	9.29	+9.60
	$\dot{Q}_{\text{loss}}$ [W]	3.25	5.25	-2.00
Test 3	ΔT_{hot} [K]	16.42	19.25	-2.83
	ΔT_{cold} [K]	21.13	10.27	+10.86
	$\dot{Q}_{\text{loss}}$ [W]	3.40	5.47	-2.07
Test 4	ΔT_{hot} [K]	16.02	18.33	-2.31
	ΔT_{cold} [K]	21.28	9.75	+11.53
	$\dot{Q}_{\text{loss}}$ [W]	3.44	5.29	-1.85

To isolate the true physical modeling error (δ_{model}) in compliance with the ASME V&V 20-2009 framework, the validation uncertainty (u_{val}) must be evaluated. In accordance with a conservative verification posture, the standard numerical discretization uncertainty was set to the maximum value identified during the Solution Verification stage, namely $u_{\text{num}} = 1.59\%$ (corresponding to the fine-grid Δp_{hot} metric).

Combining u_{num} with the standard instrumentation accuracy ($u_D = \pm 1.0^\circ\text{C}$) yields a baseline validation uncertainty $u_{\text{val},0} = \sqrt{u_{\text{num}}^2 + u_D^2}$. Evaluating the absolute comparison errors against this baseline threshold demonstrates that:

$$|E| \gg u_{\text{val},0} \quad (7.4)$$

across all operational regimes. Following the ASME V&V 20 diagnostic criteria, this condition asserts that the observed discrepancies represent a dominant signal rather than numerical discretizations or experimental measurement noise. However, this can not be consider a final conclusion because we are not considering the contribution of input uncertainties.

The dominant error observed on the cold stream (ΔT_{cold} overestimation of approximately $+9.6 \text{ K} \div +11.5 \text{ K}$) and the corresponding underestimation of thermal losses ($\dot{Q}_{\text{loss}}$ under-

estimation of $-1.85 \text{ W} \div -2.17 \text{ W}$) could be primarily attributed to unquantified input uncertainties (u_{input}). First, minor low-frequency oscillations in the experimental mass flow rate ($\dot{m}$) were recorded, whereas the numerical model enforced perfectly stationary boundary conditions. Second, the computational domain assumes smooth, ideal fluid-solid interfaces. In reality, the SLA 3D-printed core exhibits micro-geometrical surface roughness that disrupts the viscous sublayer and continuously modifies the real internal convective heat transfer coefficient (h_{int}) along the flow path, an effect compounded by local spatial variations in fluid temperature.

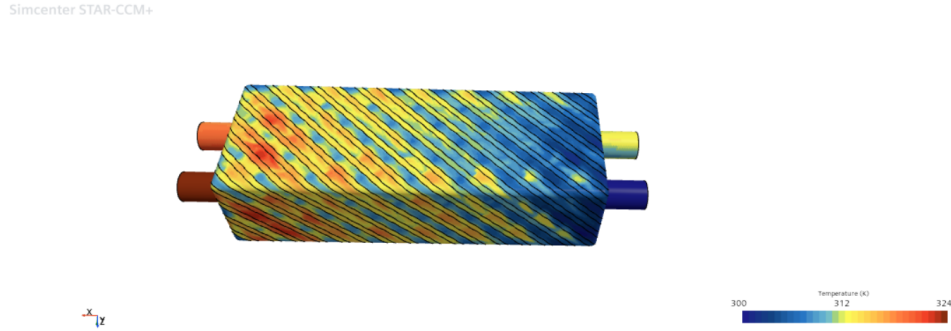


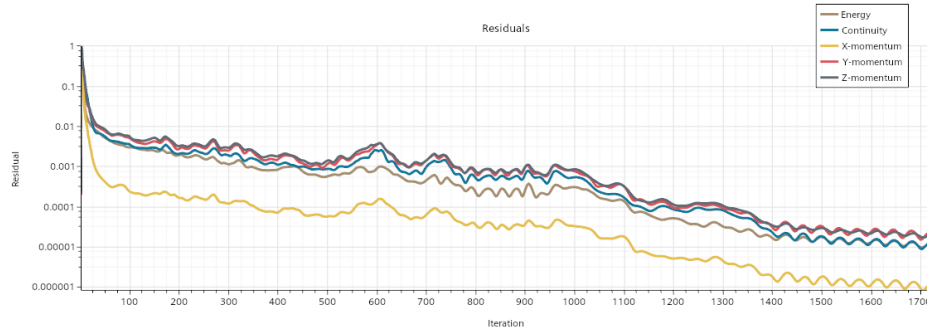
Figure 7.1: Temperature variation across wall regions (Test 1)

Crucially, the primary source of discrepancy resides in the external boundary conditions governing parasitic heat losses. The external convective heat transfer coefficient ($h_{\text{ext}} = 18 \text{ W}/(\text{m}^2\text{K})$) was prescribed based on a simplified inverse heat balance utilizing nominal mean shell temperatures. This approximation neglects spatial variations in shell temperature 7.1, non-uniform natural convection, and local thermal short-circuiting. The resulting underestimation of parasitic thermal losses directly causes the simulation to artificially retain heat within the core, thereby driving the overestimation of the cold stream temperature rise (up to $+11.53 \text{ K}$).

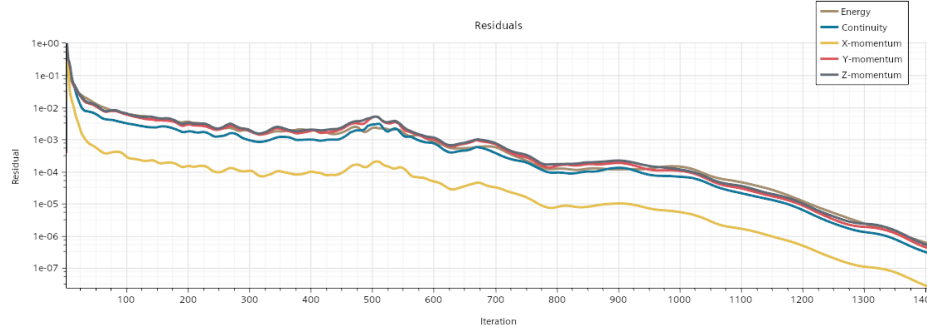
According to standard V&V methodology, when input uncertainties dominate, they must be rigorously quantified rather than treated qualitatively. Future iterations of this work must implement a formal sensitivity analysis to extract the sensitivity coefficients ($\partial S/\partial X_i$) for critical inputs like h_{ext} , $\dot{m}$, and wall roughness. By propagating these variations to formally compute u_{input} , the expanded validation interval $\delta_{\text{model}} \in [E - u_{\text{val}}, E + u_{\text{val}}]$ will correctly bound the true modeling error, clarifying whether the discrepancies arise from fundamental Navier-Stokes approximations or are simply within boundary condition noise and thermal dissipation parameters.

Nevertheless, the physical models should also be investigated further. In particular, a noticeable discrepancy exists in the residual convergence between the simulations of Test 1 and Test 4 (Figure 7.2). By the end of the computation, the residuals for Test 4 (Figure 7.2b) are almost two orders of magnitude lower than those of Test 1. Since the only difference in the entire setup is the hot stream mass flow rate (1000 l/h versus 800 l/h, respectively), a plausible explanation is that the higher flow rate pushes the hot stream into the transitional flow regime. Under these conditions, the steady laminar model struggles to resolve the emerging flow unsteadiness, preventing the residuals from dropping further. To verify this hypothesis, future studies should investigate the local Reynolds number distribution within the TPMS channels and employ transitional or

turbulent solvers to confirm whether physical unsteadiness is indeed the root cause of the residual stagnation.



(a) Residuals of Test 1



(b) Residuals of Test 4

Figure 7.2: Comparison between residuals of two different tests with hot mass-flow rate variation

7.3 Conclusions and Future Work

The thermo-hydraulic investigation of the TPMS heat exchanger successfully demonstrated the viability of additive manufacturing for ultra-compact thermal management systems with the limit of the machine itself. The correct handling and maintenance of the printer is the essential first step for a proper sample creation and experimental test. The computational framework provided valuable insight into the trade-offs between pressure drop penalties and heat transfer enhancement across different triply periodic topologies. However, more realistic conditions should be investigated, such as a higher mass flow rate to work with a turbulent flow regime and water as fluid since it is the most commonly used thermal vector for standard HX.

To further advance the development and accuracy of this methodology, the following future research directions are proposed:

1. **Turbulence and Transition Analysis:** Investigate the critical Reynolds number at which flow transition actually occurs across different TPMS topologies. This includes evaluating which turbulence models (e.g., transitional RANS closures like $\gamma-Re_\theta$ or LES) are most accurate and appropriate for capturing the specific fluid dynamic behavior of each architecture.
2. **Manifold Optimization and Inlet Effects:** Perform a parametric shape optimization of the inlet/outlet headers to ensure uniform flow distribution and mini-



mize stagnant recirculation zones. A specific focus will be placed on assessing how the inlet manifold design directly influences and contributes to the overall pressure drops across the complex internal channels.

3. **Surface Roughness in Metal TPMS:** Investigate the impact of real surface roughness on metallic TPMS components fabricated via appropriate additive manufacturing techniques (e.g., SLM). Given the high thermal conductivity of these metal parts, incorporating realistic roughness models into the numerical analysis will effectively demonstrate how manufacturing defects and partially fused powder influence not only the hydraulic penalties (Δp) but also the overall heat transfer performance.



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